

Solving Quadratic Inequations

- Solve **quadratic** inequations graphically & algebraically

Assessment:

Criteria A Test

Solving Quadratic Inequations Graphically

To solve a quadratic inequality graphically:

- Graph both sides of the inequality, or rearrange into the form $f(x) > 0$ or $f(x) < 0$ and graph.
- Determine the points of intersection of both functions, or find the zeros of the function.

To check your solution algebraically:

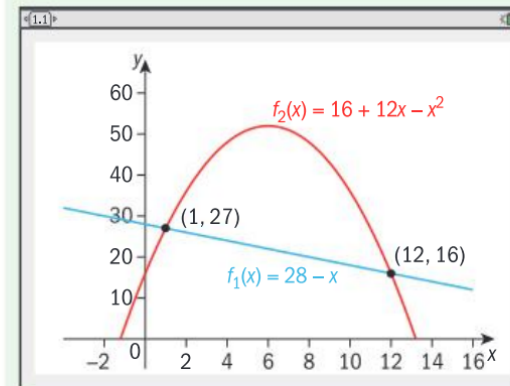
- Choose a point in the region you think satisfies the inequality, and test this point in the inequality. If it satisfies the inequality, then the x -values in this region satisfy the inequality. If your test point does not satisfy the inequality, then test the x -values in the other region(s).



Example 1

Solve the inequality $28 - x \leq 16 + 12x - x^2$. Check your solution algebraically.

Method 1



Graph both sides of the inequality and determine the points of intersection.

The x -values of points on the line (LHS) are lower than those on the quadratic (RHS) for $1 < x < 12$.

Test a point with x -value between 1 and 12:

When $x = 2$: $28 - 2 = 26$

$$16 + 12 \times 2 - 2^2 = 36$$

$$26 \leq 36 \quad \checkmark$$

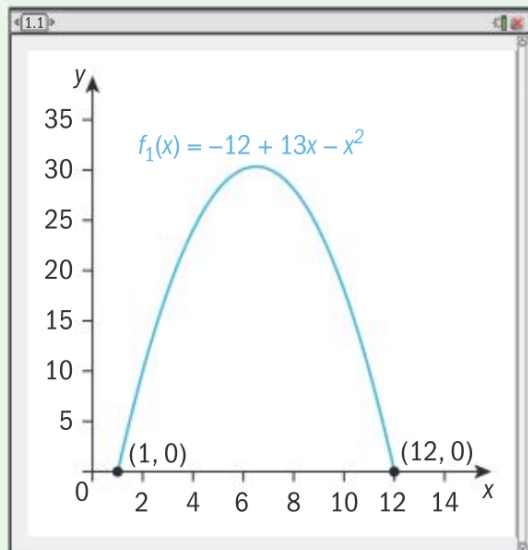
The solution is $1 \leq x \leq 12$.

Select an x -value in the solution interval, and check that it satisfies the inequality. Because the original inequality was "less than or equal to", the endpoints ($x = 1$ and $x = 12$) are included.

Method 2

$$28 - x \leq 16 + 12x - x^2$$
$$\Rightarrow 0 \leq -12 + 13x - x^2$$

Rearrange to $f(x) \geq 0$.



Graph the quadratic function.

The solution is $1 \leq x \leq 12$.

The zeros of the quadratic are $x = 1$ and $x = 12$.
 $f(x) \geq 0$ for the x -values between these zeros.

Check: When $x = 2$: $-12 + 13 \times 2 - 2^2 = 10 \geq 0$.

Select an x -value in the solution interval,
and check that the inequality is satisfied.

Question: Solve this quadratic inequality graphically

Solve the quadratic inequality $x^2 - x > 12$

Solving Quadratic Inequations Algebraically

To solve a quadratic inequality algebraically:

- Rearrange the inequality so that either $f(x) < 0$ or $f(x) > 0$.
- Factorize the quadratic and find its zeros.
- Place these two x -values on a number line, dividing it into intervals.
- Select an x -value in one of the intervals and substitute it into the original inequality. If the inequality is true, then that interval is part of the solution. If the inequality is false, then it is not.
- Repeat for all intervals.

To check your solution algebraically:

- Choose a point in the region you think satisfies the inequality, and test this point in the inequality. If it satisfies the inequality, then the x -values in this region satisfy the inequality.

Hint: Sketching a graph for quadratic inequations is often helpful !

Example:

Solve the quadratic inequality $x^2 - x > 12$ and check your answer graphically.

$$x^2 - x > 12$$

$$x^2 - x - 12 > 0$$

$$(x-4)(x+3) > 0$$

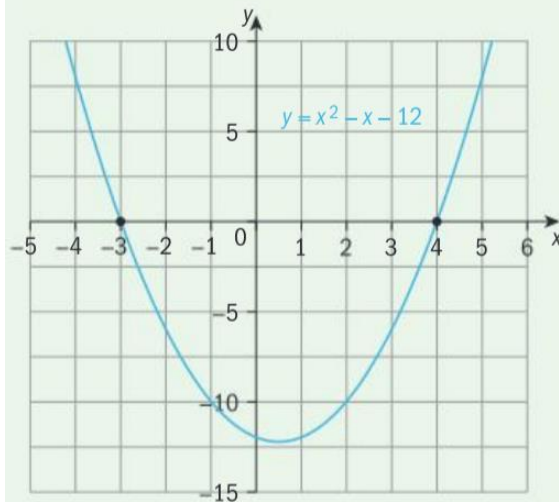
The zeros of $x^2 - x - 12$ are $x = -3$ and $x = 4$.

Interval	$x < -3$	$-3 < x < 4$	$x > 4$
Test point	-4	0	5
Substitution in $x^2 - x - 12$	$(-4)^2 - (-4) - 12 = 8$ $8 > 0$	$0^2 - 0 - 12 = -12$ $-12 < 0$	$5^2 - 5 - 12 = 8$ $8 > 0$
Is $x^2 - x - 12 > 0$?	Yes	No	Yes

From the table, the solution is $x < -3$ or $x > 4$.

Hence, $f(x) > 0$ for $x < -3$ or $x > 4$.

Check Graphically:



From the graph, you can see that both cases solve the original inequality since $x^2 - x - 12 > 0$ when $x < -3$ or when $x > 4$.

The solution is $x < -3$ or $x > 4$.

Examples:

Solving algebraically and
check solutions by
graphing

Solve for x :

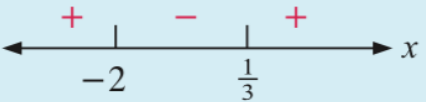
a $3x^2 + 5x \geq 2$

b $x^2 + 9 < 6x$

► Solutions

a $3x^2 + 5x \geq 2$
 $\therefore 3x^2 + 5x - 2 \geq 0$ {making RHS zero}
 $\therefore (3x - 1)(x + 2) \geq 0$ {fully factorising LHS}

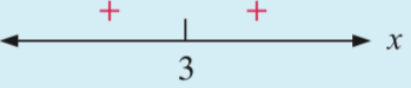
Sign diagram of LHS is



So, for LHS ≥ 0 , $x \leq -2$ or $x \geq \frac{1}{3}$.

b $x^2 + 9 < 6x$
 $\therefore x^2 - 6x + 9 < 0$ {making RHS zero}
 $\therefore (x - 3)^2 < 0$ {fully factorising LHS}

Sign diagram of LHS is



The LHS is never negative, so the inequality is **never true**.

EXERCISE 19C

1 Solve for x :

a $2x \geq 6$

d $x - 7 > -x$

b $-x < 3$

e $x + 1 \leq -2x - 4$

c $2x \leq 1 - x$

f $x^2 > 9$

2 Solve for x :

a $(x - 1)(x - 3) \leq 0$

d $(x + 5)^2 < 0$

g $x^2 < 16$

j $x^2 \leq x + 2$

m $3x^2 \geq 2(x + 4)$

p $4 < 5x^2 + 8x$

b $(2x + 3)(4 - x) > 0$

e $x^2 - 2x \geq 0$

h $3x^2 \leq 12$

k $x^2 - 4x + 4 < 0$

n $3x^2 - 6x + 3 > 0$

q $6(x^2 + 2) < 17x$

c $(x + 1)(x - 2) > 0$

f $4x^2 + 2x < 0$

i $x^2 + 4x - 5 > 0$

l $2x^2 + 7x < -6$

o $2x^2 - 5 \leq 3x$

r $9x^2 \leq 12x - 4$

Solutions:

EXERCISE 19C

- 1** **a** $x \geq 3$ **b** $x > -3$ **c** $x \leq \frac{1}{3}$ **d** $x > \frac{7}{2}$
 e $x \leq -\frac{5}{3}$ **f** $x < -3$ or $x > 3$
- 2** **a** $1 \leq x \leq 3$ **b** $-\frac{3}{2} < x < 4$ **c** $x < -1$ or $x > 2$
 d no solutions **e** $x \leq 0$ or $x \geq 2$ **f** $-\frac{1}{2} < x < 0$
 g $-4 < x < 4$ **h** $-2 \leq x \leq 2$ **i** $x < -5$ or $x > 1$
 j $-1 \leq x \leq 2$ **k** no solutions **l** $-2 < x < -\frac{3}{2}$
 m $x \leq -\frac{4}{3}$ or $x \geq 2$ **n** $x \in \mathbb{R}, x \neq 1$ **o** $-1 \leq x \leq \frac{5}{2}$
 p $x < -2$ or $x > \frac{2}{5}$ **q** $\frac{4}{3} < x < \frac{3}{2}$ **r** $x = \frac{2}{3}$